CP1.2: Argand Diagrams & Modulus-Argument Form

1. Argand Diagram Basics

Argand diagram: A 2D coordinate system where a complex number $z = a + bi$ is represented as the point (a, b). The horizontal axis is the *real axis* and the vertical axis is the *imaginary axis*. The complex number can also be thought of as the position vector from the origin to the point (a, b).

Geometrically:

- $|z|$ = distance from the origin to the point
- $\arg(z)$ = angle from the positive real axis (anticlockwise positive)
- z^* = reflection of z in the real axis
- $-z$ = rotation of z by π radians (reflection through origin)
- iz = rotation of z by $\frac{\pi}{2}$ anticlockwise

2. Modulus-Argument (Polar) Form

Any complex number z can be written in **modulus-argument form** (also called polar form):

$$z = r(\cos\theta + i\sin\theta)$$

where $r = |z| \geq 0$ and $\theta = \arg(z)$, with $-\pi < \theta \leq \pi$.

Converting Cartesian to Polar

Given $z = a + bi$:

$$r = \sqrt{a^2 + b^2}, \qquad \theta = \operatorname{atan2}(b, a) \text{ (quadrant-aware arctan)}$$

Then $z = r(\cos\theta + i\sin\theta)$.

Converting Polar to Cartesian

Given $z = r(\cos\theta + i\sin\theta)$:

$$a = r\cos\theta, \qquad b = r\sin\theta$$

Example: Write $z = -\sqrt{3} + i$ in modulus-argument form.

$r = \sqrt{3+1} = 2$
z is in the 2nd quadrant. Reference angle: $\arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$
$\arg(z) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

Therefore: $z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$

Exam tip: Leave arguments as exact values (multiples of π where possible). Only use decimals if the question says so. "Give your answer in exact form" means use surds and π.

3. Multiplying and Dividing in Polar Form

Multiplication: Multiply the moduli and add the arguments:

$$z_1 z_2 = r_1 r_2 \Big(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2) \Big)$$

Division: Divide the moduli and subtract the arguments:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\Big(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)\Big)$$

Geometric meaning: Multiplying by a complex number of modulus r and argument θ is equivalent to scaling by factor r and rotating anticlockwise by θ.

Example: If $z_1 = 4\big(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\big)$ and $z_2 = 2\big(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\big)$, find $z_1 z_2$ and $\frac{z_1}{z_2}$.

$z_1 z_2 = 8\big(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\big) = 8i$

$\dfrac{z_1}{z_2} = 2\big(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\big) = 2\Big(\frac{\sqrt{3}}{2} + \frac{1}{2}i\Big) = \sqrt{3} + i$

4. Loci on the Argand Diagram

A **locus** is the set of all points satisfying a given condition. In the Argand diagram, z represents a variable complex number and conditions on z trace out curves.

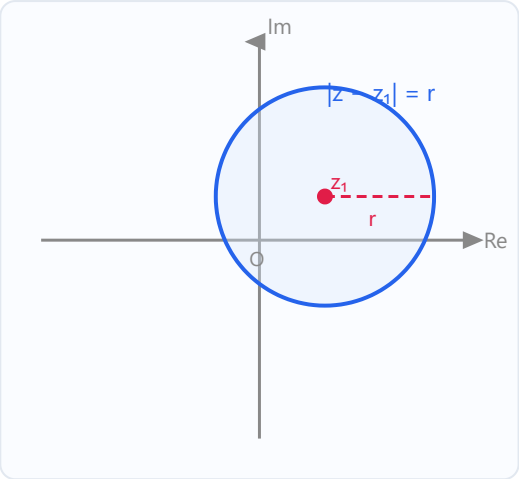
Locus Type 1: Circle — $|z - z_1| = r$

$|z - z_1| = r$ represents a **circle** with centre z_1 and radius r.

Reason: $|z - z_1|$ is the distance from the variable point z to the fixed point z_1. Setting this equal to r gives all points exactly r away from z_1.

Example: Describe the locus $|z - (3 + 2i)| = 4$.

This is a circle centred at $(3, 2)$ with radius 4.
In Cartesian form: $(x - 3)^2 + (y - 2)^2 = 16$



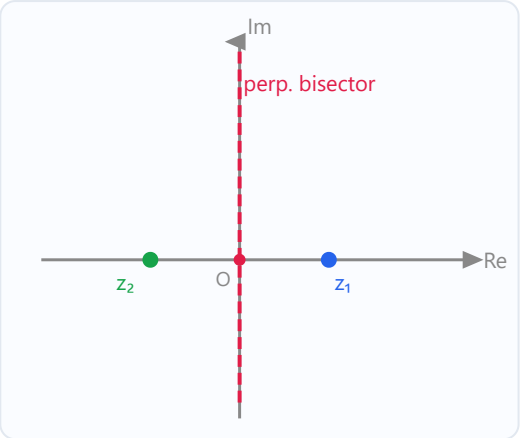
Locus Type 2: Perpendicular Bisector — $|z - z_1| = |z - z_2|$

$|z - z_1| = |z - z_2|$ represents the **perpendicular bisector** of the line segment joining z_1 and z_2.

Reason: All points equidistant from two fixed points lie on the perpendicular bisector of the segment joining them.

Example: Find the Cartesian equation of the locus $|z - 2| = |z - 6i|$.

Let $z = x + yi$.
$|z - 2|^2 = (x - 2)^2 + y^2$
$|z - 6i|^2 = x^2 + (y - 6)^2$
Setting equal: $(x - 2)^2 + y^2 = x^2 + (y - 6)^2$
$x^2 - 4x + 4 + y^2 = x^2 + y^2 - 12y + 36$
$-4x + 4 = -12y + 36$
$-4x + 12y = 32 \Rightarrow y = \frac{1}{3}x + \frac{8}{3}$



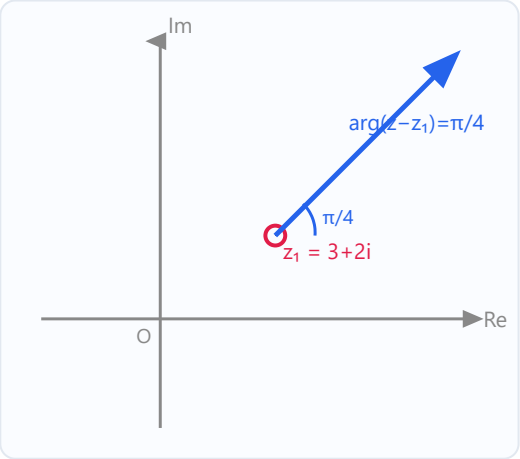
Locus Type 3: Half-Line — $\arg(z - z_1) = \theta$

$\arg(z - z_1) = \theta$ represents a **half-line** (ray) starting at z_1 (but NOT including z_1) at angle θ from the positive real direction.

Important: The point z_1 itself is excluded (it's an open endpoint). Draw a small circle at the start.

Example: Sketch $\arg(z - 3 - 2i) = \frac{\pi}{4}$.

This is a half-line starting at the point $(3, 2)$, going in the direction making angle $\frac{\pi}{4} = 45^\circ$ with the positive real direction (i.e., at gradient 1: going up-right at 45°). The starting point $(3, 2)$ is excluded.



Exam tip: For arg loci, ALWAYS state or draw that the endpoint z_1 is excluded (hollow dot). This is worth a mark and commonly dropped. Also: the locus is only a HALF-line, not the full line through z_1.

Locus Type 4: Apollonius Circle — $|z - z_1| = k|z - z_2|$, $k \neq 1$

$\dfrac{|z - z_1|}{|z - z_2|} = k$ (equivalently $|z - z_1| = k|z - z_2|$) where $k \neq 1$ and $k > 0$ represents an **Apollonius circle**.

To find the Cartesian equation, let $z = x + yi$ and square both sides after setting up the modulus expressions, then simplify into $(x - a)^2 + (y - b)^2 = r^2$.

Example: Find the locus $|z - 2| = 2|z + 1|$.

Let $z = x + yi$. Squaring both sides:
$(x - 2)^2 + y^2 = 4[(x + 1)^2 + y^2]$
$x^2 - 4x + 4 + y^2 = 4x^2 + 8x + 4 + 4y^2$
$0 = 3x^2 + 12x + 3y^2$
$x^2 + 4x + y^2 = 0$
$(x + 2)^2 + y^2 = 4$

This is a circle centred at $(-2, 0)$ with radius 2.

5. Inequalities and Regions

Replace the equality with an inequality to get a region:

- $|z - z_1| < r$: interior of a circle (exclude the boundary if strict inequality)
- $|z - z_1| > r$: exterior of a circle
- $|z - z_1| < |z - z_2|$: the half-plane on the z_1 side of the perpendicular bisector
- $\alpha < \arg(z - z_1) < \beta$: a sector (wedge-shaped region)

Example: Describe the region $|z - 3 - 4i| \leq 5$ and $0 \leq \arg(z) \leq \frac{\pi}{4}$.

Region 1: The disk (closed circle including boundary) centred at $(3, 4)$ with radius 5.

Region 2: The sector from the origin between angles 0 and $\frac{\pi}{4}$.
The required region is the intersection of these two.

Exam tip: For intersection or combined loci questions, shade the regions carefully and identify the required overlap. Mark all boundary curves and indicate whether each is included (solid line) or excluded (dashed line) depending on whether the inequality is $\leq$ or $<$.

6. Converting Between Locus Forms

The key technique for finding Cartesian equations of loci is to let $z = x + yi$ and expand:

- $|z - (a + bi)|^2 = (x - a)^2 + (y - b)^2$
- $\arg(z - (a + bi)) = \theta \Rightarrow \frac{y-b}{x-a} = \tan\theta$ (with correct half-line restriction)

Example: Find the Cartesian equation of $|z - 1 + 3i| = |z - 5 - i|$.

Let $z = x + yi$:
$|z - 1 + 3i| = |(x - 1) + (y + 3)i| = \sqrt{(x-1)^2 + (y+3)^2}$
$|z - 5 - i| = |(x - 5) + (y - 1)i| = \sqrt{(x-5)^2 + (y-1)^2}$
Squaring: $(x - 1)^2 + (y + 3)^2 = (x - 5)^2 + (y - 1)^2$
$x^2 - 2x + 1 + y^2 + 6y + 9 = x^2 - 10x + 25 + y^2 - 2y + 1$
$-2x + 6y + 10 = -10x - 2y + 26$
$8x + 8y = 16 \Rightarrow x + y = 2$

7. Summary Table of Loci

Condition	Locus	Key features
$\lvert z - z_1\rvert = r$	Circle	Centre z_1, radius r
$\lvert z - z_1\rvert = \lvert z - z_2\rvert$	Perpendicular bisector	Of segment $z_1 z_2$
$\arg(z - z_1) = \theta$	Half-line (ray)	From z_1 (excluded), angle θ

$\lvert z - z_1\rvert = k\lvert z - z_2\rvert$, $k \neq 1$	Apollonius circle	Find by letting $z = x + yi$ and squaring
$\lvert z - z_1\rvert \leq r$	Disk (region)	Interior + boundary of circle
$\alpha \leq \arg(z) \leq \beta$	Sector	Wedge from origin, include boundary lines

8. Worked Examination Style Question

Example: The complex number z satisfies both $\lvert z - 3 + i\rvert = 3$ and $\mathrm{Im}(z) \geq 0$.

(a) Sketch the locus on an Argand diagram.
(b) Find the maximum value of $\lvert z\rvert$.

Solution:
(a) $\lvert z - 3 + i\rvert = \lvert z - (3 - i)\rvert = 3$ is a circle centred at $(3, -1)$ with radius 3. With $\mathrm{Im}(z) \geq 0$, we need only the part of the circle where $y \geq 0$: the arc above the real axis, plus the two intersection points with the real axis.

(b) $\lvert z\rvert$ is the distance from the origin. Maximum is achieved at the point on the arc furthest from the origin.
Distance from origin to centre: $\sqrt{3^2 + (-1)^2} = \sqrt{10}$
Maximum $\lvert z\rvert$ = distance to centre + radius = $\sqrt{10} + 3$